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Using Event Markers

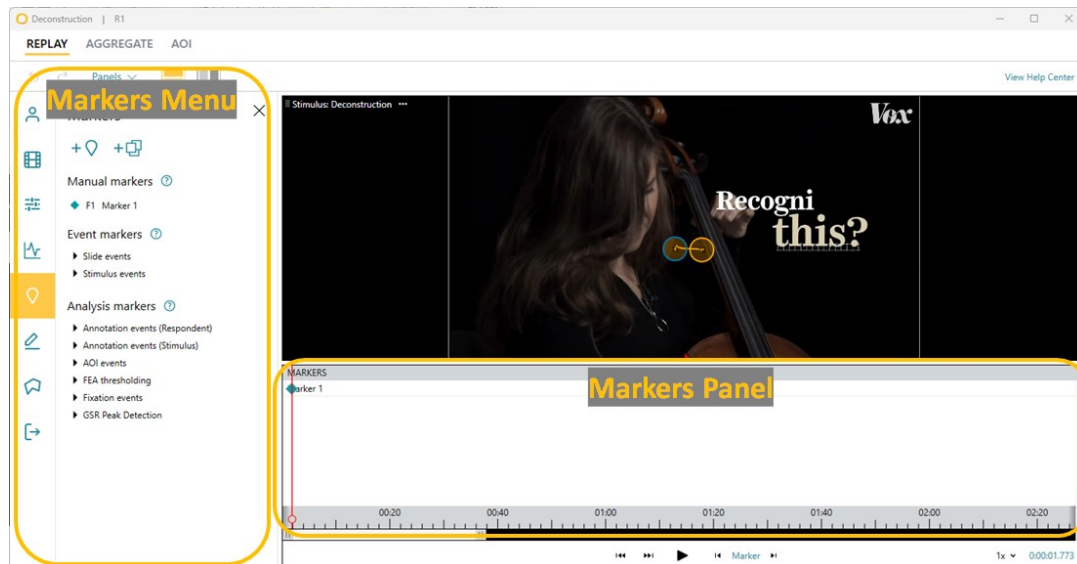
How to Create and Use Event Markers

Event Markers are used to label time points in iMotions recordings. This includes built-in iMotions events (e.g. start of recording, stimulus presentation, etc.) or a variety of user-defined events (e.g. manual events, API events, analysis events).

Event Markers can be used to create Automated Annotations to facilitate analysis.

Markers Menu and Panel

- The Markers Menu and Markers Panel can be found in any Replay window.
- The Markers Menu is used to create Manual Markers and view existing Markers (categorized by marker type).
- The Markers Panel is used to create and view Manual Marker Events alongside the recorded timeline (i.e. each instance of the Manual Marker).



Replay Window showing Annotations Menu and Annotation Panel

Types of Event Markers

Built-in Markers

Built-in markers are automatically created when they occur during recordings. Each recording contains at least one set of Slide and Stimulus events (start and stop events). Keyboard and Mouse events are created for Screen-based studies.

- Slide events indicate when any stimulus presentation starts ('StartSlide', 'StartMedia') or ends ('EndMedia', 'EndSlide')
- Stimulus events indicate when a specific stimulus is activated or deactivated (concurrent with 'StartMedia' and 'EndMedia')
- Keyboard events indicate when a specific keyboard key was pressed (only available with Screen-based studies)
- Mouse events indicate when a mouse click, scroll, or cursor movement occurred (only available with Screen-based studies)

User-Created Markers

- Live markers are created during data collection with user-inputs/Marker keys (setup in Sensor Preview)
- Manual markers are created and edited in the Markers Menu/Panel in any Replay window
- API events indicate when an event marker was received with the Event Receiving API (requiring API configuration).

Analysis Markers

- Analysis markers are created by calculating AOI metrics, which provides 'AOI dwelled on' onset/offset)
- Analysis markers are also created by running Signal Processing R-Notebooks (e.g. 'GSR peak' onset/offset).
- Calculating AOI metrics automatically creates event markers indicating when a respondent started and stopped dwelling on an AOI.
- Signal Processing R-Notebooks create event markers related to sensor signals
 - FEA Thresholding - onset/offset of passing threshold
 - GSR Peak Detection - onset/offset of GSR peaks, GSR peak (inflection point)
 - Heart Rate and HRV - R-peaks
 - Respiration Analysis - onset of inhalation and exhalation

User-Created Event Markers

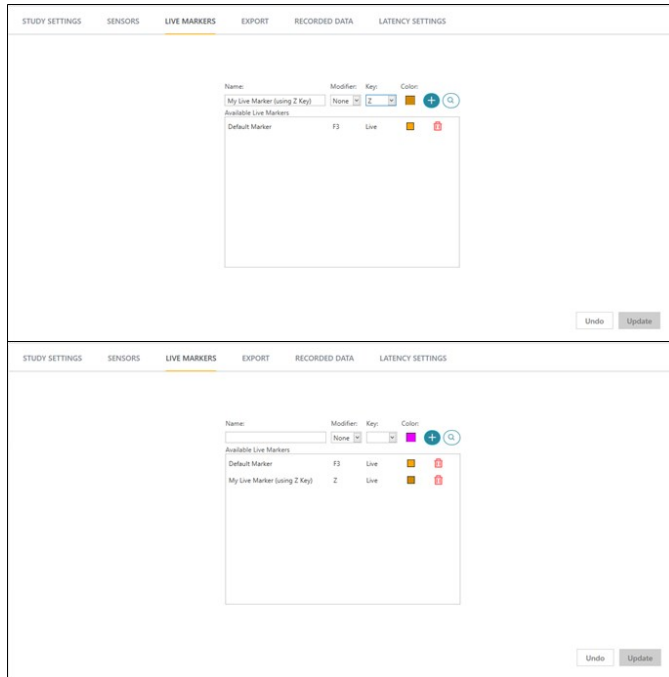
Live Markers (in Preview Window)

Live Markers are created during data collection in the Sensor Preview window, with a keyboard press (Default key: F3). Live Marker keys and labels can be further defined in [Live Marker Study settings](#).

Defining Live Markers

You have to define your Live Markers before starting the data recording. It is not possible to just press any key, you have to assign a marker to a specific key beforehand.

1. Create/Select a study.
2. Open your Study Settings 'Live Markers' tab.
3. Enter a 'Name' for your Live Marker and then allocate a Key to the Live Marker. Pressing this Key during data collection will create an Live Marker.
4. A 'Modifier' can be selected, which will also need to be pressed to create the Live Marker (Shift or Control key).
5. (Optional) Select a 'Color' to associate with the Live Marker (for display purposes).
6. Then click '+' button to set up the Live Marker.

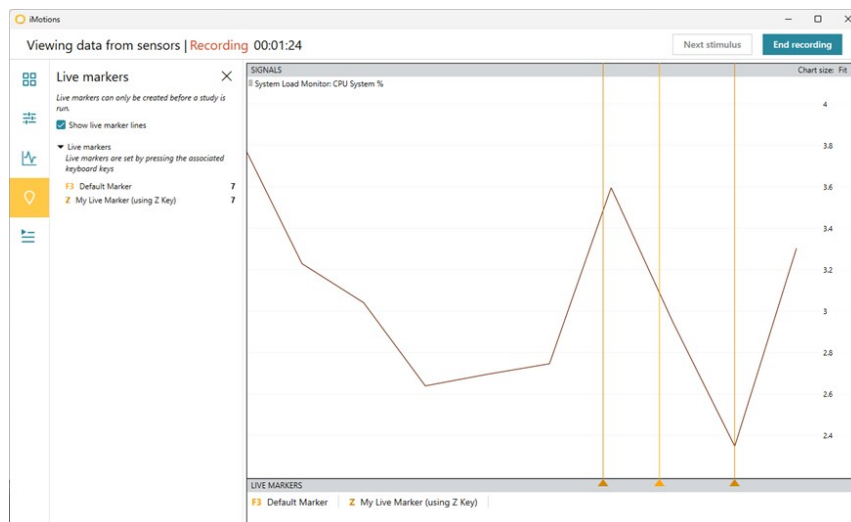


Setting up Live Markers in Study Settings

Setting Live Markers during data collection

To create Live Markers:

- During the data collection, simply hit the allocated Key, which automatically creates a Live Marker in your recording.
 - Live Markers appear in the iMotions [Sensor Preview](#). A line will appear in your Signals Panel any time a Live Marker is created.
 - The Live Markers Menu will show a list of your Live Markers with the allocated Key and number of created markers.
1. Verify 'Live Marker' Study Settings
 2. Start recording in iMotions
 3. Open the Sensor Preview window
 4. (Optional) Open 'Live markers' menu
 5. Press any Key allocated to a Live Marker

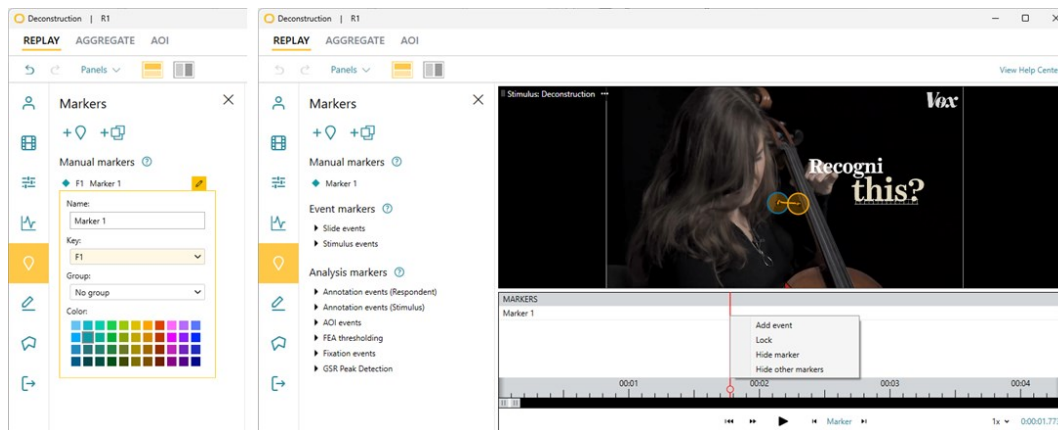


Sensor Preview showing Live Markers Menu (left) and recently created Live Markers in Signals Panel (right)

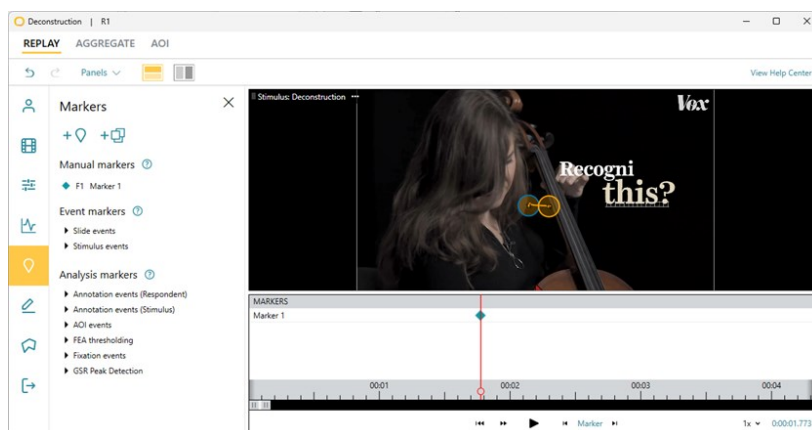
Manual Markers (in Replay Window)

To create Manual Markers:

1. Open a Replay
2. Navigate to Markers Menu
3. Click 'Add manual marker' button to create a new Manual Marker
4. (Optional) Click 'Add group' to create a group for categorizing your Manual Markers.
5. In the Markers Panel, navigate to a time point where you want to create a marker.
6. Create a marker by pressing the allocated Key (set for the Event Marker) or right-clicking and selecting 'Add event'. A maker will appear at the current time point (under the red timeline cursor)



Create markers with the Key set in your Markers Menu (left) or in the Markers Panel by right-clicking and selecting 'Add event' (right)



Manual Markers created at the current time point (under the red cursor)

Creating API Markers (from external programs)

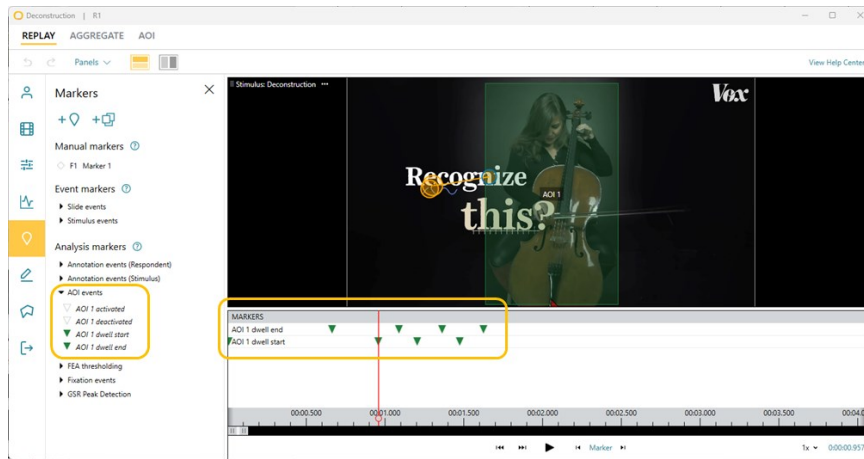
API Markers are created by the iMotions Event Receiving API, which requires a compliant event source application provided by users.

To create API markers:

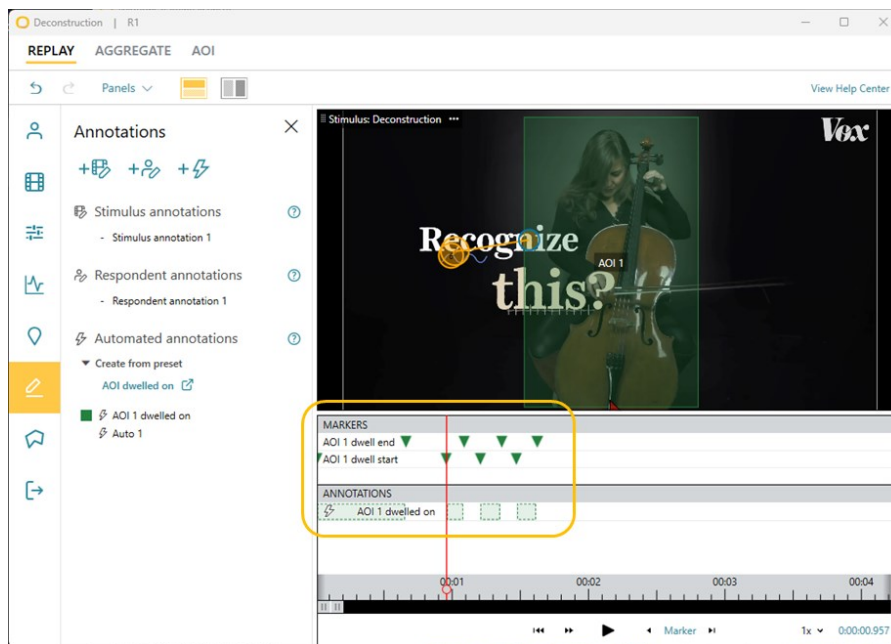
1. Set up iMotions Event Receiving API in Global Settings (API tab)
2. Set up event source application
3. Set up iMotions Study Sensors with External API Events
4. Run study in iMotions with event source application

Analysis Event Markers

Analysis Markers are automatically created by calculating AOI metrics or running Signal Processing (R Notebooks), which must be set up and performed by users.



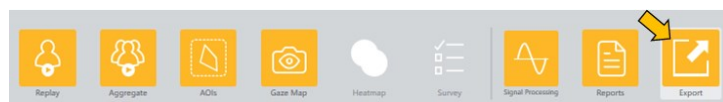
Analysis Events created by AOI Metrics Calculation



Automated Annotations based on AOI Events

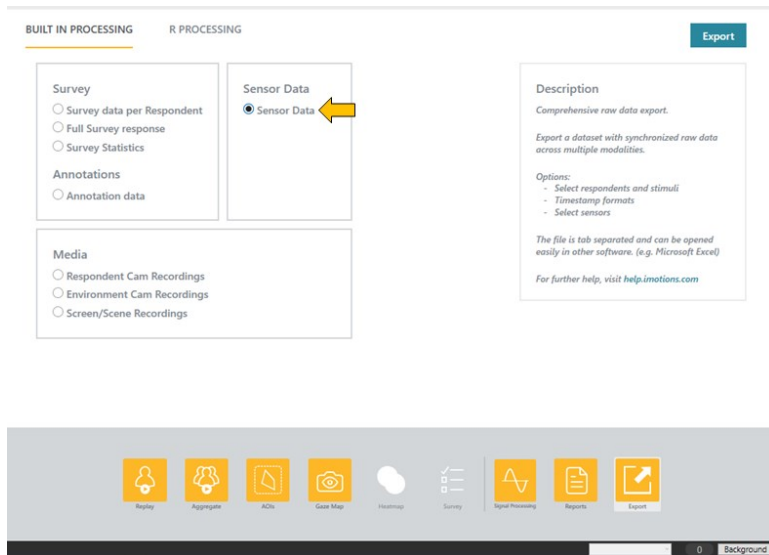
Exporting Event Markers

You can include Event Markers in Sensor Data Export files. This will create additional columns in the export file, which contain Event Marker Group, Name, and Index (count).



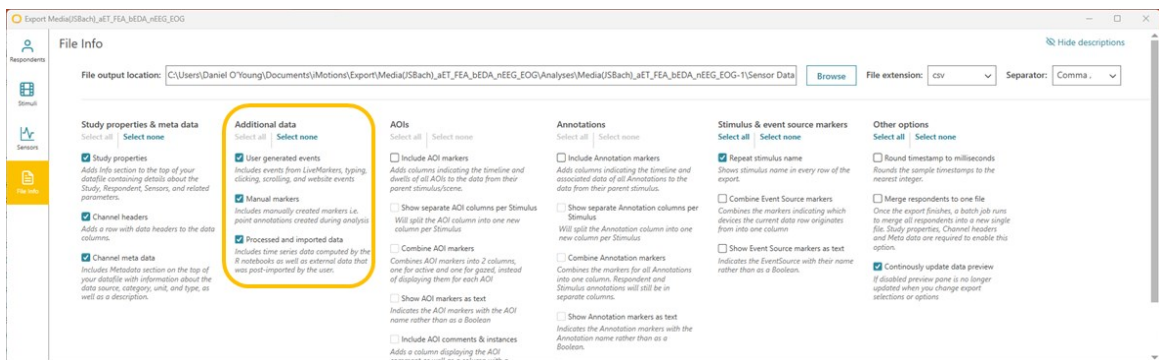
Analysis Export button

1. Click 'Export' and select Sensor Data Export



Sensor Data export

2. Click Export to open the Sensor Data Export window
3. Select the 'File Info' tab and check 'Additional data' for the Event Markers you want to include in your exported file(s). Descriptions of each setting can be displayed by clicking 'Show descriptions' (top-right corner of Export window)



'File Info' settings are used to include markers in Sensor Data export files

Scenarios for Using Event Markers

- Scenario 1: Monitoring Data Collection**
 Live Event Markers allow you to label the time points of pertinent events during data collection. This is useful for monitoring data collection, allowing the experimenter to note/classify any pertinent events as they are observed.
- Scenario 2: Coding Events in Recorded Data**
 Manual Event Markers allow you review recorded data and label time points that are relevant for analysis. This is useful when in-depth review and human-coding is needed.
- Scenario 3: Recording Events that occur in external software or hardware**
 API Event Markers allow you to create event markers with external applications that are involved in your studies, such as stimulus presentations software events (e.g. Matlab, EPrime) or hardware triggers (e.g. Cedrus Stimtracker). This allows automatic marking of these events sent from the external event source. Note: External event sources must be set up by users to comply with the iMotions Event Reviving API.
- Scenario 4: Identifying timepoints of events that were performed or detected by iMotions**
 The timepoints associated with any Event Marker can be found in any Replay (by selecting a marker) or in Sensor Data exports (based on selected 'File info' options). The export process may be helpful anytime precise timestamps (in milliseconds) are needed

